

Privacy Risk	Proposed Mitigation Solutions	NR PAPERS mentioning mitigation	RELEVANCE - high (dedicated section or implementation)	RELEVANCE - middle (only discussion)	RELEVANCE - middle COUNT	RELEVANCE - low (only brief mention)	RELEVANCE - low COUNT	RELEVANCY SCORE	NORMALIZED RELEVANCY SCORE	MATURITY -> high REPRODUCTIBILITY (code available, testing possible)	MATURITY -> high REPRODUCTIBILITY PAPERS / TOTAL NR PAPERS	MATURITY -> low REPRODUCTIBILITY (no code)	MATURITY -> high CROSS-DOMAIN GENERALITY (more than 1 dataset/domain)	MATURITY -> high CROSS-DOMAIN GENERALITY PAPERS / TOTAL NR PAPERS	MATURITY -> low CROSS-DOMAIN GENERALITY (no or only 1 dataset/domain)	MATURITY -> high DEPLOYABILITY (discussion of costs/latency)	MATURITY -> high DEPLOYABILITY PAPERS / TOTAL NR PAPERS	MATURITY -> low DEPLOYABILITY (no mention of costs/latency)	NORMALIZED MATURITY SCORE	
poa	Anonymization [10], [18], [27], [32], [41]–[43], [47], [50], [53]	10	27, 43, 53	3	18, 47	2	10, 32, 41, 42, 54	5	10.5	0.77	27, 53	0.2	10, 18, 32, 41, 42	27, 43, 53	0.3	10, 18, 32, 41, 42	43, 53	0.2	10, 18, 27, 32, 41	0.23
	Synthetic Data [33]	1	33	1	no	0	no	0	2	0.12	33	1	no	33	1	no	no	0	33	0.67
	Text Rewriting / Rephrasing [33], [37], [62]	3	62	1	no	0	33, 37	2	3	0.19	62	0.33	33, 37	62	0.33	33, 37	no	0	33, 37, 62	0.22
	Text Summarization [10], [12], [33]	3	12	1	no	0	10, 33	2	3	0.19	12	0.33	10, 33	12	0.33	10, 33	no	0	10, 33, 12	0.22
	Differential Privacy [21], [42], [43], [46], [49]	5	21, 46, 49	3	42, 43	2	42, 43	7	0.5	49	0.2	21, 42, 43, 46	21, 46, 49	0.6	42, 43	0	0	21, 42, 43, 46, 49	0.27	
	Distributed Data Storage / Hybrid Cloud Architectures [5], [55]	2	55	1	no	0	5	1	2.5	0.15	no	0	5, 55	no	0	5, 55	no	0	5, 55	0
Vector Database	Encryption [43], [50], [51]	3	43	1	no	0	50, 51	2	3	0.19	no	0	43, 50, 51	43	0.33	50, 51	43	0.33	50, 51	0.22
	Access Control [18], [22], [43], [44], [47], [50], [51], [54]	8	22, 43, 44	3	18	1	47, 50, 51, 54	4	9	0.65	22, 44	0.25	18, 43, 47, 50, 51	43	0.13	18, 22, 44, 47, 54	43	0.13	18, 22, 44, 47, 54	0.17
	Redundant Benign Knowledge Base / Duplication [25], [27], [65]	3	25, 27, 65	3	no	0	no	0	6	0.42	27	0.33	25, 65	27	0.33	25, 65	no	0	25, 27, 65	0.22
Retrieved Chunks	Synthetic Data [33]	1	33	1	no	0	no	0	2	0.12	33	1	no	33	1	no	no	0	33	0.67
	Retrieval Strategy (e.g., # of chunks, re-ranking) [12], [18], [33], [52]	4	12, 52	2	18	1	33	1	5.5	0.38	12, 52	0.5	18, 33	12	0.25	18, 33, 52	52	0.25	12, 18, 33	0.33
	Distance Metric Strategy (e.g., thresholds) [10], [12], [33]	3	12	1	no	0	10, 33	2	3	0.19	12	0.33	10, 33	12	0.33	10, 33	no	0	10, 12, 33	0.22
	Differential Privacy (in re-ranking) [86]	1	86	1	no	0	no	0	2	0.12	no	0	86	no	0	86	no	0	86	0
Answer	Local LLM Deployment [34], [43]–[45], [53], [54], [87], [88]	8	43, 45, 88	3	no	0	44, 53, 54, 87	4	8	0.58	no	0	43, 44, 53, 54, 87	43	0.13	44, 45, 53, 54, 87	43, 88	0.25	44, 45, 53, 54, 87	0.13
	Safeguards / Guardrails [45], [51], [53], [61]	4	61	1	53	1	45, 51	2	4	0.27	53, 61	0.5	45, 51	53, 61	0.5	45, 41	53	0.25	45, 51, 61	0.42
	Fact-checking / Source Citation [70]	1	no	0	no	0	70	1	0.5	0	no	0	70	no	0	70	no	0	70	0
Prompt	Anonymization (Remove/Mask/Filter PII) [10], [18], [27], [41], [50]	5	27	1	18	1	10, 41, 50	3	4.5	0.31	27	0.2	10, 18, 41, 50	27	0.2	10, 18, 41, 50	no	0	10, 18, 27, 41, 50	0.13
	Text Rewriting / Paraphrasing / Regrouping [33], [37], [40], [62]	4	40, 62	2	no	0	33, 37	2	5	0.35	40, 62	0.5	33, 37	40, 62	0.5	33, 37	no	0	33, 37, 40, 62	0.33
	Prompt Augmentation / Guardrails [25]	1	no	0	25	1	no	0	1	0.04	no	0	25	no	0	25	no	0	25	0
Backdoor	Multi-party Computation (MPC) [55]	1	55	1	no	0	no	0	2	0.12	no	0	55	no	0	55	no	0	55	0
	Anonymization [10]	1	no	0	no	0	10	1	0.5	0	no	0	10	no	0	10	no	0	10	0
	Distance Metric Strategy [10]	1	no	0	no	0	10	1	0.5	0	no	0	10	no	0	10	no	0	10	0
	Text Summarization [10]	1	no	0	no	0	10	1	0.5	0	no	0	10	no	0	10	no	0	10	0
	Knowledge Expansion [57]	1	no	0	no	0	57	1	0.5	0	no	0	57	no	0	57	no	0	57	0
Data Extraction	Detection of Anomaly Clusters [57]	1	no	0	no	0	57	1	0.5	0	no	0	57	no	0	57	no	0	57	0
	Anonymization [10], [27]	2	27	1	no	0	10	1	2.5	0.15	27	0.5	10	27	0.5	10	no	0	10, 27	0.33
	Distance Metric Strategy [10]	1	no	0	no	0	10	1	0.5	0	no	0	10	no	0	10	no	0	10	0
	Prompt Augmentation / Guardrails [40]	1	40	1	no	0	no	0	2	0.12	40	1	no	40	1	no	no	0	40	0.67
	Text Summarization [10]	1	no	0	no	0	10	1	0.5	0	no	0	10	no	0	10	no	0	10	0
Data Poisoning	Fine Tuning [60]	1	60	1	no	0	no	0	2	0.12	no	0	60	60	1	no	no	0	60	0.33
	Access Control [28], [50], [51], [67]	4	67	1	no	0	28, 50, 51	3	3.5	0.23	no	0	28, 50, 51, 67	no	0	28, 50, 51, 67	no	0	28, 50, 51, 67	0
	Anonymization [10], [18], [41], [42], [50]	5	no	0	18	1	10, 41, 42, 50	4	3	0.19	no	0	10, 18, 41, 42, 50	no	0	10, 18, 41, 42, 50	no	0	10, 18, 41, 42, 50	0
	Distance Metric Strategy [10], [30], [42]	3	30	1	no	0	10, 42	2	3	0.19	no	0	10, 30, 42	30	0.33	10, 42	no	0	10, 30, 42	0.11
	Fact-checking [70]	1	no	0	no	0	70	1	0.5	0	no	0	70	no	0	70	no	0	70	0
	Guardrails [28], [51], [61]	3	61	1	no	0	28, 51	2	3	0.19	61	0.33	28, 51	61	0.33	28, 51	no	0	28, 51, 61	0.22
	Retrieval Strategy [18], [37], [66], [72], [81]	5	66, 72	2	18	1	37, 81	2	6	0.42	no	0	18, 37, 66, 72, 81	66, 72	0.4	18, 37, 81	66	0.2	18, 37, 72, 81	0.2
	Rewriting / Rephrasing [28], [37], [38], [62], [66], [71]–[73], [81]	9	38, 62, 66, 71, 72	6	no	0	28, 37, 81	3	13.5	1	62, 73	0.22	28, 38, 37, 66, 71	38, 62, 66, 71, 72	0.67	28, 37, 81	66, 73	0.22	28, 37, 38, 62, 71	0.37
	Text Summarization [10], [78]	2	78	1	no	0	10	1	2.5	0.15	78	0.5	10	no	0	10, 78	no	0	10, 78	0.17
	Text Duplication / Knowledge Expansion [77]–[79]	3	77, 78, 79	3	no	0	no	0	6	0.42	78	0.33	77, 79	77, 79	0.67	78	no	0	77, 78, 79	0.33
	Text Filtering [19], [28], [38], [73], [77], [79]	6	19, 38, 73, 77, 79	5	no	0	28	1	10.5	0.77	73	0.17	19, 28, 38, 77, 79	19, 38, 73, 77, 79	0.83	28	73	0.17	19, 28, 38, 77, 79	0.39
	Perplexity [28], [58], [62], [72], [73], [77], [81]	7	62, 72, 73, 77, 81	5	58	1	28	1	11.5	0.85	62, 73	0.29	28, 58, 72, 77, 81	62, 72, 73, 77, 81	0.71	28, 58	73	0.14	28, 58, 62, 72, 77	0.38
	Adversarial Training [18], [20], [41], [63], [68], [69]	6	63, 68, 69	3	18, 20	2	41	1	8.5	0.62	63, 68	0.33	18, 20, 41, 69	63, 68, 69	0.5	18, 20, 41	69	0.17	18, 20, 41, 63, 68	0.33
Jailbreak	Grammar Checker [20]	1	no	0	20	1	no	0	1	0.04	no	0	20	no	0	20	no	0	20	0
LLM Extraction	Guardrails / Alignment [25], [36], [51], [89]	4	36, 89	2	25	1	51	1	5.5	0.38	89	0.25	25, 36, 51	36, 89	0.5	25, 51	89	0.25	25, 36, 51	0.33
	Differential Privacy [8]	1	no	0	no	0	8	1	0.5	0	no	0	8	no	0	8	no	0	8	0
	Anonymization [18], [41]	2	no	0	18	1	41	1	1.5	0.08	no	0	18, 41	no	0	18, 41	no	0	18, 41	0
Membership Inference	Distance Metric Strategy [10], [18]	2	no	0	18	1	10	1	1.5	0.08	no	0	10, 18	no	0	10, 18	no	0	10, 18	0
	Prompt Augmentation / Guardrails [35], [36], [39]	3	35, 36, 39	3	no	0	no	0	6	0.42	no	0	35, 36, 39	35, 36, 39	1	no	no	0	35, 36, 39	0.33
	Retrieval Strategy (e.g. re-ranking) [18], [35], [39]	3	35, 39	2	18	1	no	0	5	0.35	no	0	18, 35, 39	35, 39	0.67	18	no	0	18, 35, 39	0.22
	Prompt Rewriting / Rephrasing [35], [39]	2	35, 39	2	no	0	no	0	4	0.27	no	0	35, 39	35, 39	1	no	no	0	35, 39	0.33
	Text Summarization [10]	1	no	0	no	0	10	1	0.5	0	no	0	10	no	0	10	no	0	10	0
	Differential Privacy [83]	1	no	0	no	0	83	1	0.5	0	no	0	83	no	0	83	no	0	83	0
Prompt Extraction	Anonymization [10], [18], [50]	3	no	0	18	1	10, 50	2	2	0.12	no	0	10, 18, 50	no	0	10, 18, 50	no	0	10, 18, 50	0
	Prompt Augmentation / Guardrails [56]	1	56	1	no	0	no	0	2	0.12	56	1	no	56	1	no	56	1	no	1
	Prompt Rewriting / Rephrasing [56]	1	56	1	no	0	no	0	2	0.12	56	1	no	56	1	no	56	1	no	1
Prompt Injection	Guardrails [25], [41], [51]	3	no	0	25	1	41, 51	2	2	0.12	no	0	25, 41, 51	no	0	25, 41, 51	no	0	25, 41, 51	0
	Text Filtering [31]	1	no	0	no	0	31	1	0.5	0	no	0	31	no	0	31	no	0	31	0
	Grammar Checker [31]	1	no	0	no	0	31	1	0.5	0	no	0	31	no	0	31	no	0	31	0

[illegible]